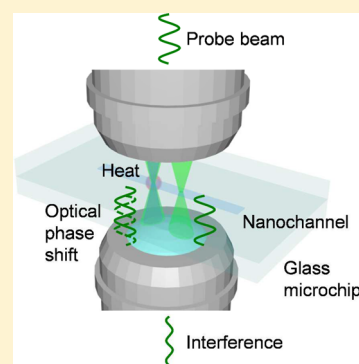


Thermo-optical Characterization of Photothermal Optical Phase Shift Detection in Extended-Nano Channels and UV Detection of Biomolecules

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ABSTRACT: The expansion of microfluidics research to nanofluidics requires absolutely sensitive and universal detection methods. Photothermal detection, which utilizes optical absorption and nonradiative relaxation, is promising for the sensitive detection of nonlabeled biomolecules in nanofluidic channels. We have previously developed a photothermal optical phase shift (POPS) detection method to detect nonfluorescent molecules sensitively, while a rapid decrease of the sensitivity in nanochannels and the introduction of an ultraviolet (UV) excitation system were issues to be addressed. In the present study, our primary aim is to characterize the POPS signal in terms of the thermo-optical properties and quantitatively evaluate the causes for the decrease in sensitivity. The UV excitation system is then introduced into the POPS detector to realize the sensitive detection of nonlabeled biomolecules. The UV-POPS detection system is designed and constructed from scratch based on a symmetric microscope. The results of simulations and experiments reveal that the sensitivity decreases due to a reduction of the detection volume, dissipation of the heat, and cancellation of the changes in the refractive indices. Finally, determination of the concentration of a nonlabeled protein (bovine serum albumin) is performed in a very thin 900 nm deep nanochannel. As a result, the limit of detection (LOD) is 2.3 μM (600 molecules in the 440 attoliter detection volume), which is as low as that previously obtained for our visible POPS detector. UV-POPS detection is thus expected to be a powerful technique for the study of biomolecules, including DNAs and proteins confined in nanofluidic channels.



There has been rapid growth in the field of microfluidics, which integrates micrometer-scale structures on a centimeter-scale substrate and controls liquid samples.^{1,2} The characteristics of microspaces, including reduced sample volume and reaction time due to a high surface-to-volume ratio and fast mass transfer, has led to the realization of high performance, rapid, and low cost analyses, which has triggered innovation in analytical chemistry. More recently, much smaller spaces at the 10–1000 nm scale (extended-nano space) have gathered attention for the development of novel functions using the unique characteristics of the extended-nano space.^{3,4} In particular, the electrophoretic separation of DNA molecules utilizing the geometries of nanostructures,^{5,6} chromatographic separation utilizing the extremely high surface-to-volume ratios of nanochannels,^{7,8} and concentration polarization⁹ and ionic current rectification¹⁰ utilizing the overlap of electric double layers in nanospaces have been reported. The analyses of stretched DNA confined in nanochannels¹¹ and proteins using surface-dominant spaces¹² have also been implemented intensively. However, the sample volumes in extended-nano spaces are from attoliters (aL) to femtoliters (fL). Therefore, the number of analyte molecules is no more than 10^3 , even for concentrations from micromolar to millimolar. Thus, detection methods for extended-nano spaces are required to be absolutely sensitive.^{13,14}

Our group has previously proposed a methodology for microfluidic integration that combines continuous flow

chemical processing (CFCP)¹⁵ and thermal lens microscopy (TLM).^{16–19} The concept of CFCP consists of the following three steps. First, a complicated analytical procedure is divided into simple unit operations, such as mixing, reaction, and separation. Second, each unit operation is miniaturized using microfluidic techniques. Third, the micro unit operations (MUOs) are connected similar to an electrical circuit to integrate the entire chemical process. Finally, the product or analyte is detected by TLM using heat generated via optical absorption and nonradiative relaxation of the target molecule. In microfluidics, laser-induced fluorescence (LIF) is frequently used as a sensitive detection technique,²⁰ although LIF requires fluorescent labeling of analytes because few molecules have fluorescence. On the other hand, TLM can be used to detect almost all molecules that have optical absorption because most molecules release energy of absorbed photons as heat. Furthermore, the largest advantage of photothermal detection techniques is high sensitivity, even at the single particle/molecule level.^{21–23} In particular, TLM can detect a concentration that corresponds to 97 pM (0.4 molecules in 7 fL).²⁴ Although a weakness of TLM is the fact that it detects a wide range of molecules in complex samples, selective and sensitive analyses can be realized by combining preparative

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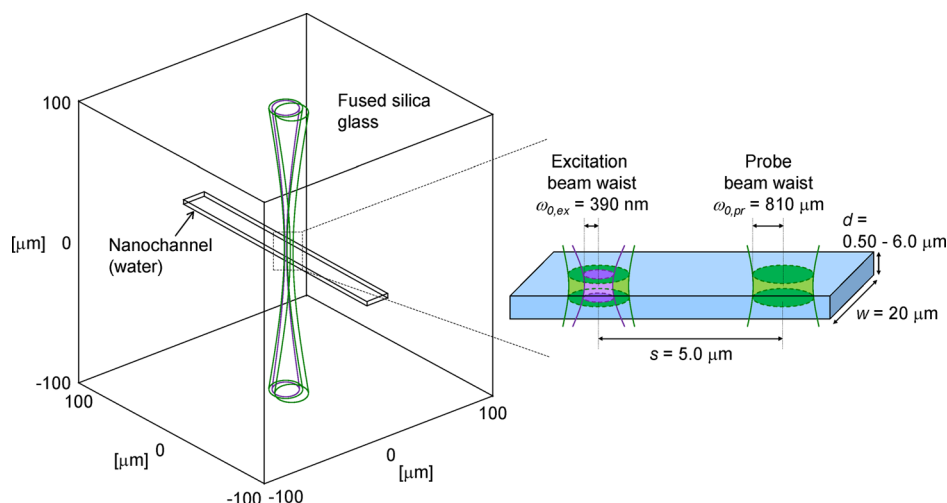


Figure 1. Geometry of the nanochannel and the excitation and probe beams used for the finite element analysis.

treatments, reactions and separation by CFCP and detection by TLM. The integration methodology of CFCP is useful in nanofluidics as well as microfluidics. Nevertheless, conventional TLM is difficult to apply to nanofluidics because conventional TLM detects a change in refractive index because of a temperature increase via the lensing effect with a thermal lens. Although geometrical optics can be used to describe the same refraction by the thermal lens when the size of the thermal lens is much larger than the wavelength of the light, it is no longer useful in nanochannels that are smaller than the wavelength. For this reason, we have developed a novel technique to detect the photothermal optical phase shift (POPS) through the introduction of a differential interference contrast (DIC) method to the TLM technique.²⁵ The principle of POPS detection is based on heat generation from the analyte and the interferometric detection of a phase shift of the probe beam because of the change in the refractive index. This principle, which is described by the wave optics, makes detection of the POPS effective, even if the size of the nanochannel is smaller than the wavelength. To date, the POPS detector, which was previously called DIC-TLM, has been verified in principle and has realized the sensitive determination of dye molecule concentrations in nanochannels,²⁶ and the detection of separated molecules by chromatography using an extended-nano channel.²⁷ However, ultrahigh sensitivity at the single particle/molecule level has not been achieved for POPS detection in nanochannels. One reason is that the solvent was water, which has high thermal conductivity and low temperature gradient of refractive index. For example, Tokeshi et al. used benzene, which has low thermal conductivity and a high temperature gradient of refractive index to enhance the photothermal signal.²⁴ Gaiduk et al. used glycerol to prevent heat transfer to the glass substrate because it has low thermal conductivity.²² Glycerol also has high viscosity, which may suppress diffusion of single analyte molecules under the laser focus and thus provide a very long signal integration time to enhance the sensitivity. However, the use of these solvents is unrealistic for nanofluidic experiments, especially in biomedical applications. Another reason is the decrease in sensitivity specific to nanochannels that may be caused by following three reasons. First of all, the decrease in sensitivity is related to the ultrasmall detection volume determined by the sizes of the nanochannels. Second, the dissipation of the heat generated in

the ultrasmall space to the surrounding media is unavoidable. Third, the refractive index of the analyte solution decreases with an increase in the temperature, while that of the glass nanochannel increases. Therefore, the changes in the refractive indices of water and glass may cancel each other out and result in decreased sensitivity. These three effects have been suggested previously; however, their specific contributions have not been evaluated quantitatively. In addition, the previous POPS detector was limited to target molecules that have optical absorption in the visible wavelength region, which made the range of application narrow. Therefore, to realize application of the POPS detector to the analysis of biomolecules including DNAs, proteins, and peptides, requires the introduction of UV excitation.

In the present study, our primary aim was to investigate the thermo-optical characteristics of the POPS signal and the causes of the decrease in sensitivity specific to nanochannels. In addition, we have aimed to develop a new POPS detector with UV excitation. The new POPS detector was designed and constructed for UV lasers using a symmetric microscope system, a UV objective lenses, and tailor-made DIC prisms. After verification of the principle, the sensitivity for POPS detection was characterized with respect to the size of the nanochannels. The experimental results were then compared with those from a simulation by the finite element analysis method. Finally, the concentration of nonlabeled protein molecules was determined using the UV-POPS detector with consideration of the thermo-optical characteristics.

■ EXPERIMENTAL SECTION

Principle. The principle of POPS detection has been explained elsewhere.²⁵ Briefly, a probe laser beam is separated into two beams and mixed again by a pair of DIC prisms installed in a microscope. An excitation laser beam that is not separated by a DIC prism is focused onto one of the probe beam spots. The excitation beam is absorbed by the analyte molecules in the sample solution and heat is released via nonradiative relaxation. A local change in the refractive index is then induced in the sample solution, which produces a phase shift for one of the probe beams. The phase shift is finally detected by interference of the probe beams as a change of the probe beam intensity.

To detect the POPS effectively, one of the probe beams must transit the center of the heat source and the other probe beam must transit the region without heat. Therefore, the shear value, which is the distance between the two parallel probe beams, should be designed to match the size of the photothermal effect. The size of the photothermal effect is estimated from the thermal diffusion length. The thermal diffusion length in water is approximately $5\ \mu\text{m}$ when the modulation frequency of the excitation beam is $2\ \text{kHz}$.

Simulation. Finite element analysis of the thermal diffusion and the POPS signal in nanochannels was performed to investigate the thermo-optical characteristics. Figure 1 shows the 3D geometry of a nanochannel, and the probe and excitation beam profiles used for the simulation. The source term $Q(r, z, t)$ of the heat conduction equation due to the focused excitation beam is described as follows.^{28,29}

$$Q(r, z, t) = \frac{1}{2} \frac{4P\alpha}{\pi^2 \omega_{\text{ex}}^2} \exp(-\alpha z) \exp\left(\frac{-2r^2}{\omega_{\text{ex}}^2}\right) [1 + \cos(2\pi f t)] \quad (1)$$

$$\omega_{\text{ex}}^2 = \omega_{0,\text{ex}}^2 \left(1 + \frac{z^2}{z_{r,\text{ex}}^2}\right) \quad (2)$$

Here, $P\ [\text{W/m}^2]$ is the excitation power, $\alpha\ [\text{m}^{-1}]$ is the absorption coefficient, $f\ [\text{Hz}]$ is the modulation frequency, $\omega_{0,\text{ex}}\ [\text{m}]$ is the beam waist radius, and $z_{r,\text{ex}}\ [\text{m}]$ is the Rayleigh length of the excitation beam. After solving the heat conduction equation in the time-dependent mode, the change in refractive index multiplied by the intensity of the focused probe beam is integrated as follows.³⁰

$$S(r, t) = \int_z \left(\frac{dn}{dT}\right) (T - T_0) \left(1 + \frac{z^2}{z_{r,\text{pr}}^2}\right) dz \quad (3)$$

Here, $S(r, t)$ is the integral of the change in refractive index detected by the probe beam, $(dn/dT)\ [-]$ is the temperature coefficient of the refractive index,³¹ $T(r, z, t)\ [\text{K}]$ is the calculated temperature, $T_0\ [\text{K}]$ is the initial value of the temperature, and $z_{r,\text{pr}}\ [\text{m}]$ is the Rayleigh length of the probe beam. The (dn/dT) of fused silica was taken from the technical data sheet provided by the supplier (Shin-Etsu Chemical, Tokyo, Japan). The POPS signal $\Delta S(t)$ was then calculated as the phase contrast between the two probe beams as follows:

$$\Delta S(t) = S(r = 0\ \mu\text{m}, t) - S(r = 5.0\ \mu\text{m}, t) \quad (4)$$

The heat source term has a component that oscillates with the modulation frequency, so that $\Delta S(t)$ also oscillates at the same frequency. Therefore, the amplitude of the oscillation is the POPS signal that is detected by a lock-in amplifier. All simulations and calculations were performed using COMSOL Multiphysics 4.3b.

Design. The new microscope designed for UV-POPS detection is illustrated in Figure 2a. The microscope has a symmetrical structure that consists of two UV objective lenses and custom-made DIC (Nomarski) prisms, while the visible POPS detector has an asymmetrical structure with an objective lens, condenser lens, Nomarski prism, and Wollaston prism.²⁵ The symmetrical microscopic system was easily designed and is expected to have a higher precision of interference with the use of two identical DIC prisms. The microscope and DIC prisms

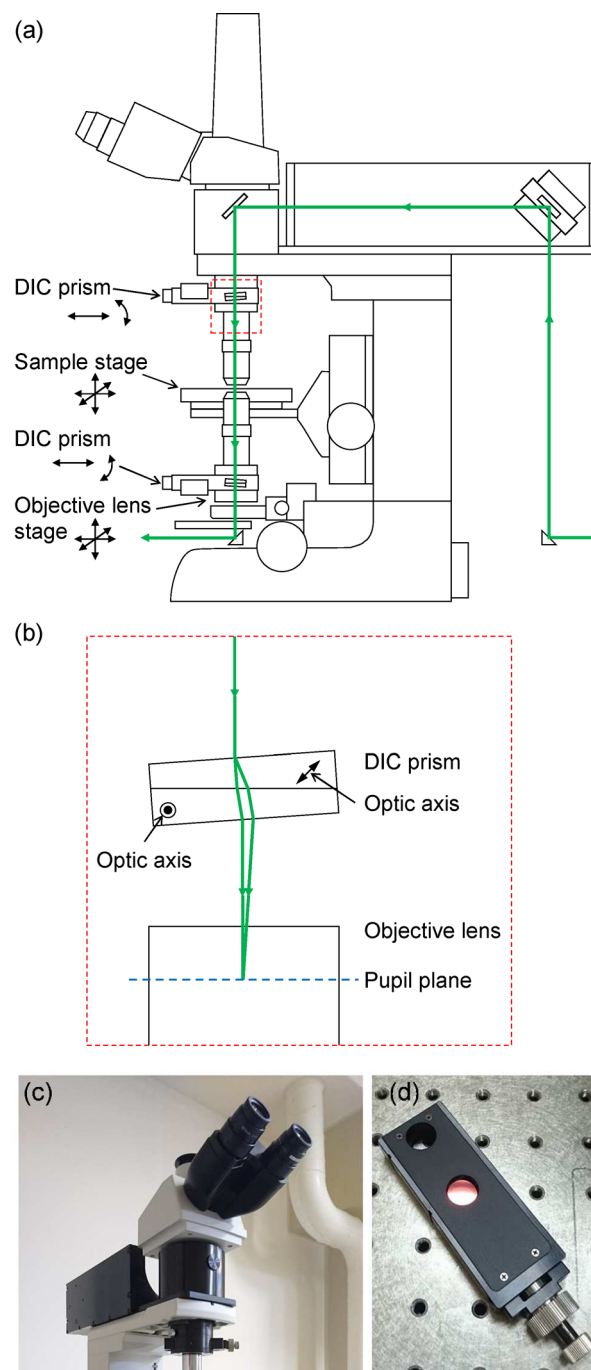


Figure 2. (a) Design of the UV-POPS detector. The green line indicates the light path of the probe beam. (b) Design of the DIC prism. The angles of the probe beams may be emphasized. (c, d) Photographs of (c) the UV-POPS detector and (d) the DIC prism.

have separate micro adjustment mechanisms to achieve highly precise interference.

The shear values of the DIC prisms were determined as $5\ \mu\text{m}$ to match the thermal diffusion length in water. To produce two parallel probe beams under the objective lenses, the plane that the two probe beams cross must be matched to the pupil plane of the objective lens, as illustrated in Figure 2b. The DIC prisms were designed and fabricated to satisfy the shear value and split angle using a lithium tetraborate crystal ($\text{Li}_2\text{B}_4\text{O}_7$), which has high transparency and birefringence in the UV region. The DIC prisms were also fabricated using an optical contact without an adhesive agent that would be subject to degradation by the high-power UV laser. Figure 2c and 2d shows photographs of the microscope and the DIC prisms.

Setup. Figure 3 shows the entire optical system of the UV-POPS detector. The optical system is almost the same as that

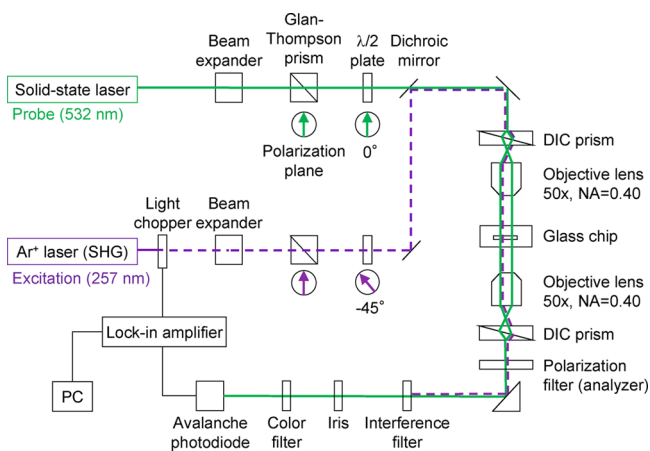


Figure 3. Optical systems of the UV-POPS detector. The green and purple arrows indicate the polarization planes of the probe and excitation beams, respectively.

used for our visible excitation POPS detector, except for the microscope, and the probe and excitation beams. The probe and excitation beam were produced by a solid-state laser (532 nm, Compass 315M, Coherent, Santa Clara, CA, USA) and from the second harmonic of an Ar^+ laser (257 nm, INNOVA 300C FreD, Coherent), respectively. The wavelengths were selected because the UV objective lenses (UV 50 $\times$, NA = 0.40, Nikon Engineering, Tokyo, Japan) were designed to minimize chromatic aberration for 532 and 266 nm.³² The excitation and probe spots were precisely aligned in the x , y , and z directions at a micrometer level using reflection mirrors and beam expanders. The invisible UV excitation spot was visualized using riboflavin solution. The power of the excitation beam was 7 mW under the objective lens. The separation of the excitation and probe beams was controlled by changing their polarization planes. The excitation beam was intensity-modulated at 2.5 kHz by a mechanical chopper and cut by interference and color filters. The interfering probe beam was extracted using a slit and detected using an avalanche photodiode (APD) module (C12703, Hamamatsu Photonics, Hamamatsu, Japan). The output signal of the APD was processed by a lock-in amplifier (LI5640, NF Corporation, Yokohama, Japan).

Fabrication. Fused silica chips with extended-nano channels were fabricated using electron-beam lithography (EBL), photolithography and reactive-ion etching (RIE).³³ The nanochannels were fabricated on a fused silica substrate

($30 \times 70\ \text{mm}^2$, VIOSIL-SQ, Shin-Etsu Chemical, Tokyo, Japan) using EBL and RIE. Guide microchannels were fabricated on another substrate using photolithography and RIE. After drilling connection holes on the microchannels, the substrates were bonded using a low-temperature bonding machine.³⁴ The width of all nanochannels was $20\ \mu\text{m}$ and the depths were 500 nm, 900 nm, $2\ \mu\text{m}$, and $6\ \mu\text{m}$. The width and depth of all the microchannels were 500 and $6\ \mu\text{m}$, respectively.

RESULTS AND DISCUSSION

Verification of Principle. The extinction ratio of the polarized probe beam was first measured to ensure that the separated probe beams were interfering correctly. The extinction ratio was 1.3×10^{-2} , which was almost the same as that obtained for the visible POPS detector and sufficient to perform POPS detection.²⁵ The extinction ratio is much larger than that obtained with a commercial DIC microscope and prisms, which is attributed to the long UV objective lenses that have many optical components inside to degrade the polarization. The dependence of the POPS signal on the excitation beam polarization was investigated next. The POPS signal was detected only when the excitation beam was not separated (polarization angle = $\pm 45^\circ$). When both the excitation and probe beams were separated (polarization angle = 0° or 90°), the signal was minimized, and no phase contrast was detected. The dependence of the signal on the polarization is clear evidence of POPS detection, which is never observed in other photothermal detection techniques.

Sensitivity in Extended Nanochannels. In contrast to microchannels, the sensitivity of the POPS detection in extended nanochannels is largely dependent on the channel size because the size of extended-nano channel is smaller than the diffraction limits and confocal parameters of the laser beams. In such cases, the detection volume and the absolute number of detected analyte molecules must be discussed, rather than the concentration of the analyte. Figure 4 shows calibration curves for an aqueous solution of adenine obtained

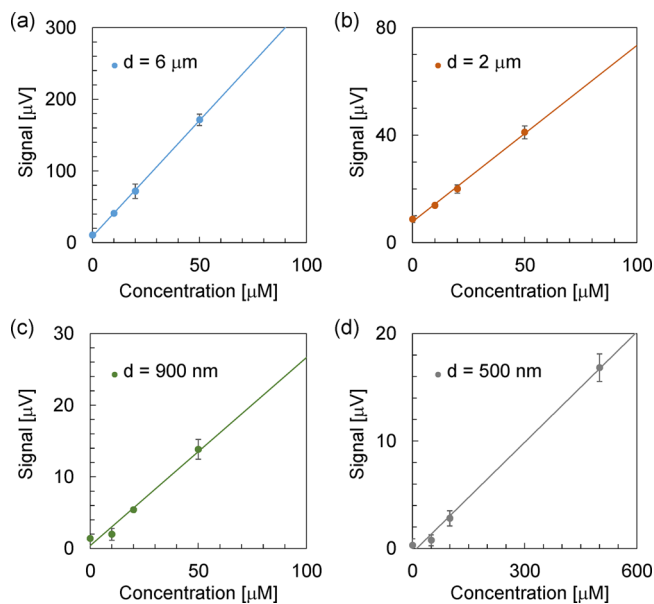


Figure 4. Calibration curves for adenine aqueous solution in $20\ \mu\text{m}$ wide nanochannels with depths of (a) $6\ \mu\text{m}$, (b) $2\ \mu\text{m}$, (c) 900 nm, and (d) 500 nm.

using four channels with the same channel width ($20\ \mu\text{m}$) and different channel depths ($6\ \mu\text{m}$, $2\ \mu\text{m}$, $900\ \text{nm}$, and $500\ \text{nm}$). The sensitivity, that is, the slope of calibration curve, decreased monotonically with the channel depth. One reason for the decrease of sensitivity is simply the reduction in the detection volume. The detection volume for UV-POPS detection is estimated from the radius of the excitation beam waist and the channel depth, as illustrated in Figure 1. As the channel depth decreases, the detection volume and the absolute number of analyte molecules decrease, which leads to the decrease in sensitivity.

Thermo-optical Characterization of POPS Signal.

However, the sensitivity decrease cannot be fully explained by the reduction in the detection volume (detection volume loss), because the heat released inside the nanochannel quickly dissipates into the glass (thermal loss). Furthermore, the change in the refractive index of the glass is positive with an increase in temperature ($dn/dT > 0$), while the change in the refractive index of water is negative ($dn/dT < 0$), which may cause a cancellation of the changes in the refractive indices (optical loss) as suggested in our previous paper.²⁶ Figure 5a

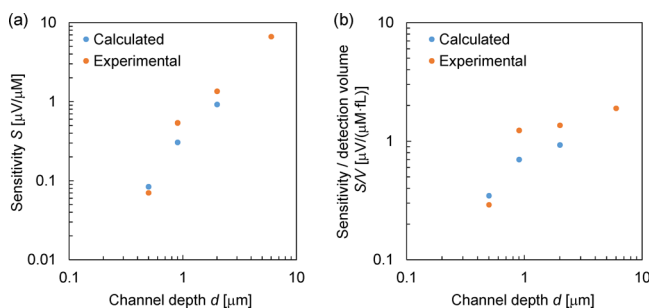


Figure 5. Comparison of the POPS sensitivities obtained by finite element analysis and experiment. (a) Sensitivity and (b) sensitivity per detection volume.

shows the sensitivities obtained by simulation and experiment plotted as a function of the channel depth. The calculated and experimental results show a similar tendency of sensitivity decrease, which supports that the thermo-optical characterization of the POPS signal is valid. The following calculations were performed to discuss the influences of the detection volume loss, thermal loss, and optical loss separately. First, the detection volume and the loss of detection volume (relative value compared to a $6\ \mu\text{m}$ channel depth) were calculated for each channel depth from Gaussian beam profiles. Second, the changes in refractive indices for water and glass were calculated separately. The POPS signal generated by the change in the refractive index of water reflects the detection volume and thermal loss, while excludes the optical loss. Third, the total POPS signal, which includes the change in the refractive index of glass, was calculated to consider the detection volume, and both the thermal and optical losses. Thus, each component of signal loss was evaluated separately by subtraction from the calculated POPS signals. Figure 6 summarizes the contributions of the detection volume loss, thermal loss, and optical loss over the total POPS signal. When the channel depth was $6\ \mu\text{m}$, the contribution of the change in the refractive index of glass (optical loss) was only 2.7%, if the change in the refractive index of water was assumed to be 100%. Thus, the total POPS signal was 97.3%. Even in the $6\ \mu\text{m}$ deep channel, the heat generated in the nanochannel was partly dissipated, although

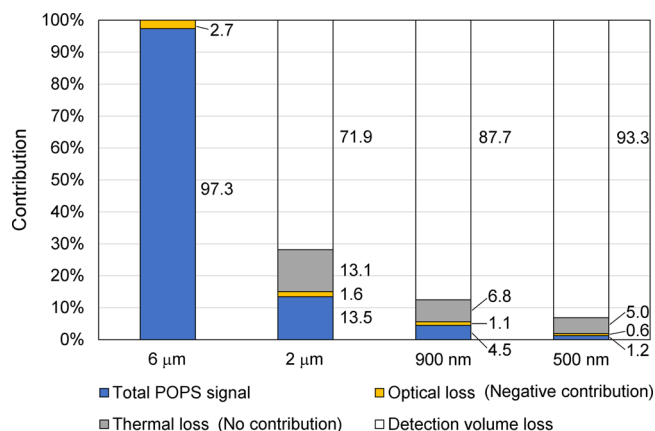


Figure 6. Contributions of the detection volume, thermal loss, and optical loss to the total sensitivity decrease for POPS detection in micro/nanochannels.

the thermal loss is not shown in the column. When the channel depth was $500\ \text{nm}$, the detection volume was only 6.7% compared to that for the $6\ \mu\text{m}$ deep channel, which indicates a detection volume loss of 93.3%. However, the POPS signal calculated from the change in the refractive index of water was only 1.7% compared to that for $6\ \mu\text{m}$ deep channel; the difference (5.0%) was regarded as thermal loss. Furthermore, the total POPS signal was decreased to 1.2% due to the optical loss (0.6%). Figure 6 shows that the detection volume loss, which is inevitable for measurements in extended-nano channels, has the largest contribution to the decrease in sensitivity. Therefore, the sensitivity per detection volume was evaluated and is shown in Figure 5b. The experimental result showed that the sensitivity, corrected with respect to the detection volume, decreased rapidly at $900\ \text{nm}$.

Detection of Nonlabeled Protein. Determination of the concentration of a nonlabeled protein was performed using a $900\ \text{nm}$ deep nanochannel. Figure 7 shows a calibration curve

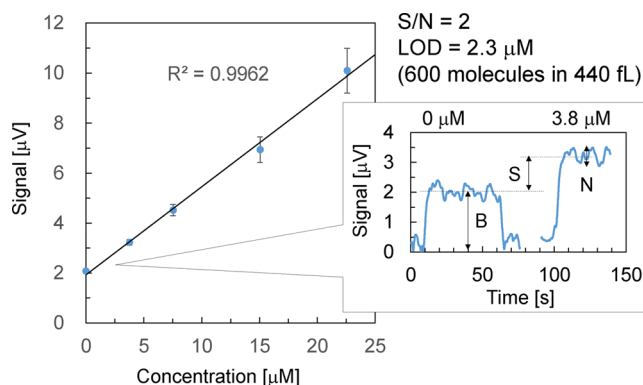


Figure 7. Calibration curve for BSA in PBS. The LOD was calculated from an S/N of 2.

for bovine serum albumin (BSA) in a phosphate buffer saline (PBS) solution. One measurement was performed for 30 s and the mean values and standard deviations of three times measurements are plotted in the graph. The time constant of the lock-in amplifier was 1 s. The limit of detection (LOD) was calculated from the results for low concentration samples as a concentration that gives a signal-to-noise ratio (S/N) of 2, as shown in the inset of Figure 7. The calculated LOD was 2.3

μM , which corresponds to 600 BSA molecules in the detection volume of 440 aL. This LOD is comparable to that with our visible POPS detector (2.4 μM , 390 molecules in 250 aL) obtained using dye molecules, which has a similar absorption coefficient to that of BSA.²⁶ Therefore, it was concluded that the UV-POPS detector can realize the same level of performance as the visible POPS detector. These LOD appears to be lower than that in the previous reports (e.g., 97 pM, 0.4 molecules in 7 fL).²⁴ This is due to the following three reasons: First, a metallic porphyrin in benzene was used as a sample in the previous study. The absorption coefficient of BSA ($\epsilon = 22000 \text{ M}^{-1} \text{ cm}^{-1}$ at 257 nm) is smaller than that of the porphyrin ($\epsilon = 32000 \text{ M}^{-1} \text{ cm}^{-1}$ at 488 nm).^{35,36} Moreover, benzene can enhance the photothermal sensitivity by 31 times because of its low thermal conductivity and high temperature gradient of refractive index.³¹ Second, the sensitivity decreases with decreasing detection volume, as discussed earlier. The detection volume reported here is 16 times smaller than that of the previous report. Here we also discuss the decrease in sensitivity due to thermal diffusion. From Figure 6, the contribution of the total POPS signal (4.5) is smaller than that of the thermal loss (6.8), which suggests that more than half of the heat generated in the nanochannel is lost as thermal loss. By multiplying these factors, the potential sensitivity in the present work is less than 10^{-3} times lower than that in the previous study. Thus, the LOD for BSA in PBS is reasonable considering the solvent properties and the disadvantageous thermal conditions in the nanochannel.

CONCLUSION

To conclude, the thermo-optical characteristics of POPS detection in nanochannels was investigated by simulation using the finite element analysis method and by experiment using a UV excitation POPS detector. The UV-POPS detector was designed based on a symmetrical microscope system with two DIC prisms and UV objective lenses to separate and mix the visible probe beam with a shear value of 5 μm . The DIC prisms were fabricated to have high transparency and durability for the UV excitation beam. The UV-POPS detector was verified to cause interference of the probe beams with a low extinction ratio (1.3×10^{-2}). The dependence of the POPS signal on the polarization of the excitation beam was verified as previously conducted with the visible POPS detector. The simulation and experimental results showed that the sensitivity for POPS detection was significantly decreased with the detection volume, moderately decreased by heat dissipation to the nanochannel material, and slightly decreased by cancellation of the changes in the refractive indices. This is the first quantitative evaluation of the decrease in the sensitivity of photothermal measurements in nanochannels and is a key development toward more sensitive measurements, including single molecule detection. Finally, the sensitive detection of BSA protein was performed in a very thin 900 nm deep nanochannel and the LOD was 2.3 μM (600 molecules in a detection volume of 440 aL). Thus, high performance comparable to visible light excitation was achieved using UV excitation. The introduction of UV excitation is expected to broaden the range of applications of POPS detection to nanofluidics. Separation techniques for DNAs, amino acids, and peptides using nanofluidics are expected to progress significantly with the use of UV-POPS. In addition, UV-POPS has a significant advantage for analyses with extremely small sample

volumes, such as femtoliter sampling from a single cell, with which fluorescent labeling is difficult.³⁷

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Notes

The authors declare no competing financial interest.

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